TITLE OF THE DISSERTATION IN ALL CAPITAL LETTERS

by

Candidate Full Name

DISSERTATION

Submitted in partial satisfaction of the requirements for degree of

DOCTOR OF PHILOSOPHY

in

Biomedical Sciences

in the

GRADUATE DIVISION

of the

UNIVERSITY OF CALIFORNIA, SAN FRANCISCO

Approved:

Dissertation Chair	Date
Committee Member	Date

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DEDICATION

Dedicated to my family, mentors, and colleagues whose encouragement made this work possible.

ACKNOWLEDGEMENTS

I express my deepest gratitude to my advisor, dissertation committee members, and collaborators.

CONTRIBUTIONS

Descriptions of student and co-author contributions for adapted manuscript chapters.

EPIGRAPH

"Equipped with his five senses, man explores the universe around him and calls the adventure Science."

— Edwin Powell Hubble (1948)

ABSTRACT

COMPUTATIONAL EVALUATION OF EPIGENETIC CLOCKS AND TRANSCRIPTIONAL REGULATION ACROSS DIVERSE POPULATIONS

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Epigenetic clocks provide a molecular readout of chronological and biological age. This dissertation investigates cross-ancestry prediction performance, fine-maps methylation quantitative trait loci, and establishes standardized computational workflows for equitable biomarker translation.

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LIST OF ABBREVIATIONS

CpG	Cytosine-phosphate-Guanine dinucleotide
DNA	Deoxyribonucleic Acid
GWAS	Genome-Wide Association Study
meQTL	Methylation Quantitative Trait Loci
SNP	Single Nucleotide Polymorphism
UCSF	University of California, San Francisco

CHAPTER 1

EPIGENETIC CLOCK PREDICTION PERFORMANCE ACROSS DIVERSE POPULATIONS

1.1 Epigenetic clocks and biological aging dynamics

Biological aging is characterized by systematic molecular remodeling across the epigenome. The pioneer pan-tissue clock developed by Horvath [1] demonstrated that DNA methylation status at 353 CpG dinucleotides accurately tracks chronological age across diverse human tissues. Similarly, Hannum and colleagues [2] identified a blood-specific methylation predictor consisting of 71 markers that captured clinical markers of physiological decline.

1.1.1 Second-generation clocks and mortality prediction

Subsequent investigations introduced biomarkers trained on physiological phenotypes and mortality risks rather than purely chronological time. Specifically, PhenoAge [3] and GrimAge [4] demonstrated superior predictive validity for morbidity and all-cause mortality. However, recent cross-population benchmarks [5] revealed notable calibration disparities when applied across ancestrally diverse cohorts. Furthermore, Gopalan et al. [6] highlighted divergence in age-associated methylation dynamics between global populations.

1.1.2 Methodological harmonization across cohorts

To address platform discrepancies, we evaluated quality control filters where **probe failure rates exceeded 15% & sample heterozygosity** was systematically audited. In particular, *cross-reactive probes and non-specific hybridization* were masked prior to regularized elastic net fitting.

> Biological age acceleration represents the mathematical divergence between chronological time and physiological vitality. Epigenetic markers provide an empirical substrate to map this divergence across human lifespans, integrating genetic variation, environmental exposures, and social determinants into an actionable, quantitative metric of organismal resilience.

Table 1.1: Multi-ancestry DNA methylation profiling cohorts and technology platforms.

Cohort Name	**Ancestral Group**	**Sample Size (N)**	**Platform**
Discovery Cohort	Multi-Ancestry	621	EPIC BeadChip
Validation Cohort A	African American	1,394	450K BeadChip
Validation Cohort B	European	729	450K BeadChip

Deep learning implementations. Novel neural network architectures have shown promising accuracy for high-dimensional methylation modeling as described by Chen and colleagues [7]. Preprints on population stratification further underscore this need [8].

1.1.3 Discussion and future perspectives

Our findings underscore the imperative of ancestry-inclusive training datasets to ensure equitable clinical utility of molecular aging biomarkers.



Figure 1.1: Conceptual diagram of multi-ancestry clock evaluation and meQTL mapping. High-throughput DNA methylation profiles are filtered, evaluated for biological age acceleration, and integrated with genotype arrays for ancestry-stratified fine-mapping.

CHAPTER 2

MAPPING GENETIC ARCHITECTURE OF METHYLATION QUANTITATIVE TRAIT LOCI

2.1 Genetic architecture of epigenetic regulation

Genetic variants exert profound regulatory influences on local and distal DNA methylation levels. Previous comprehensive mapping initiatives [5] demonstrated that methylation quantitative trait loci (meQTL) are enriched in disease-associated regions. Linking these variants to aging markers [1] establishes mechanistic pathways connecting sequence variation to epigenetic age acceleration. Complementary approaches are discussed in [6].

2.1.1 Ancestry-stratified fine-mapping procedures

We implemented Bayesian fine-mapping algorithms to dissect causal variant credible sets across ancestrally distinct cohorts. Consistent with [7], trans-ancestry fine-mapping substantially narrowed the credible set sizes by exploiting differential linkage disequilibrium architecture.

> Deciphering the causal interplay between sequence-level genetic variation and dynamic epigenetic modifications is foundational to interpreting GWAS discoveries and constructing biologically robust, multi-ancestry risk prediction models.

2.1.2 Colocalization and functional validation

Colocalization analyses confirmed shared genetic etiology between meQTL and complex traits in over 42% of tested genomic loci.

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